

TRILOKESH RANJAN SARKAR

Data Analyst | Generative AI | Data Engineering | Machine Learning | Big Data

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SUMMARY

Data Analyst with 2 years of experience in Big Data, Python, SQL, Machine Learning, Generative AI, NLP, OCR, and Data Visualization. Experienced in delivering AI-driven analytics, automation, and data solutions across Retail, Healthcare, and BFSI domains. Proven track record in developing LLM-powered workflows, OCR/NLP automation, customer analytics, and data validation systems, reducing manual effort by 80% and improving operational efficiency. Skilled in data analysis, dashboard development, predictive analytics, and business intelligence using Power BI, Looker Studio, SQL, and Python to generate actionable insights and support strategic decision-making. Passionate about applying Machine Learning, Generative AI, and scalable data solutions to solve complex business challenges and drive measurable impact.

EXPERIENCE

Data Analyst

Web Spiders

10/2024 - Present Kolkata, West Bengal

- Built an LLM-powered taxonomy audit pipeline using GPT-4o, Apache Airflow, AWS, Redshift, PostgreSQL, and Label Studio to automate retail taxonomy validation, ontology auditing, web research, AI analysis, and human-in-the-loop review workflows, improving data quality, governance, and operational scalability.
- Developed OCR/NLP-based automation solutions for data extraction, handwritten document digitization, Named Entity Recognition (NER), and intelligent text parsing, significantly improving data accuracy, processing efficiency, and reducing manual effort by 80%.
- Utilized SQL, Power BI, and data visualization techniques to analyze trends, build interactive dashboards, and generate actionable business insights supporting strategic planning, reporting, and decision-making.
- Utilised pre-trained models for sentiment analysis of in-store customer facial expressions, helping track emotions and boost engagement.
- Built heat maps from video analytics to spot high-traffic areas and peak hours, improving store layout and staff planning.
- Created a custom PyTesseract handwriting OCR pipeline to digitize handwritten medical documents. Additionally, developed OCR workflows that reduce human intervention in the data extraction process by 80% by using natural language processing (NLP) for named entity recognition (NER) and regex matching.
- Developed an Automatic Number Plate Recognition (ANPR) system using YOLOv8, Computer Vision, and Deep Learning for real-time, high-accuracy license plate detection and recognition.
- Designed marketing analytics dashboards to monitor email campaign KPIs including open rate, response rate, and conversion metrics, driving data-driven optimization that improved engagement by 15% and conversions by 12%.
- Tech Stack: SQL, Python, Power BI, AWS Cloud Server, Regex, NLP.

Research Associate – Generative Artificial Intelligence

Praxis Business School

12/2023 - 05/2024 Kolkata, West Bengal

- Tested adversarial robustness of a Generative AI by comparing FGSM and Carlini-Wagner attacks to evaluate model security against adversarial threats.
- Used the Carlini-Wagner and FGSM attacks to evaluate the security of the model. Improved the robustness and efficiency of the model through optimized LLM fine-tuning under resource constraints.

EDUCATION

Post-Graduation Program in Data Science

Praxis Business School

2023 - 2024 Kolkata, West Bengal

CQPI
6.2 / 8

Bachelors of Computer Application

B. P. Poddar Institute of Management and Technology

2020 - 2023 Kolkata, West Bengal

CGPA
8.8 / 10

PUBLICATION

Comparing FGSM, Carlini-Wagner Attacks, and the Function of Distillation as a Defense Mechanism to Assess Adversarial Robustness

arXiv.org

2023 - 2024

https://arxiv.org/abs/2404.04245

- Published in Arxiv Tech's Research Gate after peer review by top GenAI researchers, recognizing the work's contribution to adversarial defense techniques in machine learning.
- Provided a thorough evaluation framework comparing model vulnerabilities and defense effectiveness, offering actionable insights for secure model deployment..

CERTIFICATIONS

Databricks Fundamentals Accreditation

Databricks

SQL Certificate

Hacker Rank

Generative AI Fundamentals Accreditation

Databricks